MUlti-country MABM

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1 Extendend Abstract

In this paper we present a basic MUlti-country Macroeconomic Agent-Based Model (MU-MABM). We consider a simplified multi-country setting in which two countries, A and B, interact only through *international trade*, i.e. the exchange of internationally tradable goods (*tradables* for short).[1] The exchange rate is irrevocably fixed and normalized to one (i.e. the two countries belong to a currency union). For the moment, we will assume complete symmetry, i.e. the parameters of the model are uniform across countries.

In each country, the economic system consists of households, firms and banks interacting on the markets for goods, labor and credit. In each country firms belong to one of two sectors: sector 1 produces tradables and sector 2 produces non-tradables or domestic goods. We assume that firms belonging to the tradable sector (T-firms for short) sell goods on the same international market irrespective of the country of residence. Firms belonging to the non-tradable sector or domestic firms (D-firms), on the contrary, can sell only to consumers residing in the same country. Therefore there are three markets for goods: an international market for tradables and two national markets for non-tradables. Labour and credit markets are domestic: in each country firms (both in sector 1 and sector 2) demand domestic labor and domestic credit.

Households supply labor in their own country[2] and consume the goods produced by both sectors. Firms in sector 1 use both labor and non-tradables as inputs, while firms in sector 2 use only labor. Therefore, non-tradable goods may be used as (i) input for the tradable sector or (ii) consumption goods for the households. Firms therefore are linked in a supply chain. A firm in sector 2 is a supplier of intermediate inputs (upstream firm) to firms in sector 1 (downstream firms). Upstream and downstream firms may reside in different countries. Hence when international trade concerns two firms (in the non tradable sector and tradable sector of different countries) we have an international supply chain. The model allows us to highlights the interaction mechanisms between the two countries.

In each country a simple financial system supplies credit to the production sectors. The interaction between the firms and the financial system is going to produce endogenous crisis due to credit crunches and positive demand feedback. Depending on the degree of integration of the two countries, the crisis will spread to the other country.

[1] For simplicity, there are no capital flows in the present version of the model.

[2] The labor input, therefore is not mobile. In other words, by assumption labor markets are domestic.